

Coding project - Campus Tech doc

Campus Pulse – Technical Specification (No-Code View)

1. High-Level Architecture

- Client: React Native (Expo) mobile app, iOS & Android
- Backend: Firebase serverless stack (Firestore, Anonymous Auth, Cloud Functions)
- Data Transport: HTTPS REST (Firestore SDK)
- Privacy Guardrails: Foreground-only location access; no background code paths; no photo or personal identifiers stored

1. Core Functional FeaturesA. Anonymous Authentication

- Firebase Anonymous Auth creates a per-device UID at first launch
- UID reused across sessions; no email, no social login
- Token refreshed automatically by SDK

B. Foreground Location Sensing

- Expo Location requests "when-in-use" permission on first launch
- Location stream starts only while the app is in the foreground and stops on background/quit
- Raw coordinates never leave the device; only a distance check (≤ 30 m) against predefined campus pins is performed
- If GPS accuracy > 15 m, check-in buttons stay disabled until accuracy improves
- If permission denied, user may still open a manual picker list of campus places

C. Place Proximity Engine

- A static GeoJSON list (~20 campus POIs) ships with the app
- Each POI contains lat/lon, type tag (dining, truck, library), and a nominal capacity integer
- Distance computed with haversine formula every location update
- When the device enters a 30 m radius of a POI, the app labels the user "at place" and surfaces the check-in UI

D. Check-In Submission

- UI presents three emoji buttons (😞😐😄) representing 0, 1, 2
- One tap writes a single document to the check-ins collection
- Fields: placeId, level (0-2), server-side timestamp, anonymous uid
- Local 90-minute block: after submission the buttons disable for that POI to reduce spam
- Write is queued offline by Firestore persistence and uploads automatically when network returns

E. Real-Time Crowd Score (Client View)

- Every POI document carries a computed busyPercent (0-100) and lastUpdate timestamp
- Client listens to onSnapshot for the places collection → map markers change color instantly
- Color mapping: 0-39 % green, 40-79 % yellow, 80-100 % red
- If lastUpdate >90 min old, marker shows desaturated color plus "stale" badge

F. Aggregation & Decay (Server Side)

- A scheduled Cloud Function runs every 5 minutes
- Loads only the last 90 minutes of check-ins
- Applies exponential decay weight: half-life = 30 minutes
- Calculates weighted average level, normalises to 0-100, writes busyPercent back to the POI document
- Old check-ins are ignored (not deleted, but fall outside the 90-minute window)

G. Offline-First Support

- Firestore persistence layer caches reads & writes on-device
- Map colours remain usable without network; sync indicator shows when data is queued
- No extra code required beyond enabling persistence flag at startup

H. Spam & Abuse Controls

- Client-side 90-minute cooldown per POI
- Cloud Function ignores any check-in older than 90 minutes or from a UID that exceeds 20 writes in 5 minutes (configurable)
- Security rules enforce field shape and value ranges (level must be 0-2)

I. Privacy & Data Retention

- No background location permission declared; manifest omits `ACCESS_BACKGROUND_LOCATION`
- No personally identifiable information stored (no name, email, phone, photo)
- Location fixes reside in RAM only; never written to disk or to Firestore
- Check-ins are logically expired after 90 minutes; a separate janitor function (optional) may delete them eventually

J. Observability & Analytics

- Firebase Analytics events auto-logged: `app_open`, `checkin_tap`, `permission_response`
- Crashlytics enabled for crash reports
- No user-specific tracking IDs beyond the anonymous uid

1. Non-Functional Features

- Build System: Expo Application Services (EAS) – cloud build for IPA/APK
- Updates: Over-the-air updates via EAS Update (no store resubmission for copy changes)
- Testing Strategy: Jest unit + React Native Testing Library component tests; GitHub Actions for CI

- Accessibility: Touch targets ≥ 48 dp; emoji buttons accompanied by accessibility labels; map supports system zoom controls
- Localization: Strings externalised for easy en/es switch (only English shipped in MVP)
- Store Compliance: Apple "Data Safety" and Google Play "Data security" sections filled with "location – precise – ephemeral – collected but not shared"

1. Feature-to-Backend Mapping Summary

- Anonymous Auth → Firebase Auth
- Foreground location → Expo Location module (client only)
- Proximity trigger → client distance check against static POI list
- Check-in write → Firestore collection checkins
- Real-time crowd score → Firestore collection places (updated by Cloud Function)
- Offline queue → Firestore persistence (built-in)
- Scheduled aggregation → Cloud Function (cron every 5 min)